

Data sources: control vs type 2 diabetes, human adipose transcriptomics

Summary of the project data manifest and alternative sources. Generated 2026-10-05.

1. Datasets in data/manifest.csv

Dataset	Study	Relevance	Files	~MB
GSE342773	Visceral adipose tissue single-nucleus RNA-seq with T2D-related groups	Direct control-versus-T2D relevance	4	320
GSE268904	Human adipose single-cell/nucleus transcriptomics study	Control-versus-metabolic disease relevance	1	998
GSE129363	Single-cell transcriptomics of human adipose stromal vascular fraction	Foundational human adipose reference cohort	5	114
GSE278526	Human adipose depot single-nucleus RNA-seq study	Control-versus-metabolic disease relevance	2	450
GSE249089	Single-nucleus RNA-seq of subcutaneous adipose tissue from 84 METSIM participants	Strong metabolic-trait and T2D relevance	4	73
GSE176171	A single-cell atlas of human and mouse white adipose tissue	Human adipose reference atlas; obesity-focused	4	766
GSE176067	Characterization of the stromal vascular fraction of human subcutaneous adipose tissue	Non-diabetic reference cohort	1	90
GSE281356	Human subcutaneous and visceral adipocyte atlases	Depot comparison reference	1	446
HRA002549	Single-cell RNA-seq of visceral adipose stromal vascular fraction in lean, obesity and T2D	Direct lean-control and T2D comparison	1	436
GSE141432	Obese insulin-resistant humans with compensatory hyperinsulinemia (abdominal subcutaneous adipose RNA-seq)	Bulk RNA-seq; lean vs obese insulin-resistant vs obese T2D	2	1
GSE71416	Gene expression in omental adipose tissue of morbidly obese diabetic African Americans	Bulk array; 14 obese T2D vs 6 obese non-diabetic omental adipose	2	50

Sizes for GSE141432 and GSE71416 are estimates and should be corrected after download. GSE71416 compares obese T2D with obese non-diabetic, not lean controls.

2. Download status

In the cloud session, www.ncbi.nlm.nih.gov was reachable but GEO file downloads redirect to <ftp.ncbi.nlm.nih.gov>, and figshare.com was blocked (403, organization policy). No data files were downloaded. Downloads should be run locally, or after allowing <ftp.ncbi.nlm.nih.gov> and figshare.com under Network access: python `scripts/download_manifest.py`

3. Alternative sources (control vs T2D / metabolic state)

Accession	Tissue	Groups	Type
GSE141432	Subcutaneous	Lean, obese insulin-resistant, obese T2D	RNA-seq (in manifest)
GSE71416	Omental	14 obese T2D vs 6 obese non-diabetic	Array (in manifest)
GSE179455	Epicardial	Obese, diabetic, lean	RNA-seq
GSE29231	Visceral	Females with T2D (controls unconfirmed)	Array

Accession	Tissue	Groups	Type
GSE20950 / GSE15773	Omental, subcutaneous	Insulin-sensitive vs insulin-resistant obese	Array
GSE133099	Mature adipocytes	Lean, obese, T2D (6 each)	Processed only; raw withheld

Other hosts (not verified from the session): CELLxGENE collections [ba84c7ba-8d8c-4720-a76e-3ee37dc89f0b](#) and [fe0e718d-2ee9-42cc-894b-0b490f437dfd](#) (h5ad), Broad Single Cell Portal SCP1376 and SCP3409, [adiposetissue.org](#), [matkp.org](#), and EGA [EGAD00001010195](#) (controlled access).